

Earnings-Call Sentiment Analysis & Return Prediction

Dual-LLM Extraction | FinBERT Speaker Sentiment | GBM Modeling

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Metric	Value
Best Model	GBM (depth=2, top-5 features)
OOS Accuracy	59.6%
Rank IC	0.142
ROC-AUC	0.595
PnL Sharpe (5D L/S)	1.08
Hit Rate	59.6%
Cum. Excess Return	+40.8%
Universe	14 tickers, 131 earnings calls
Period	Q4-2023 through Q1-2026

Table of Contents

- 1. Methodology & Pipeline Overview
- 2. Stage 1: Transcript Parsing
- 3. Stage 2: Dual-LLM Event Extraction
- 4. Stage 3: FinBERT Speaker-Level Sentiment
- 5. Stage 4: Feature Engineering (13 Features)
- 6. Target Design & Train/Test Split
- 7. Model Comparison & Results (8 Models)
- 8. Backtest: GBM 5D Long/Short Strategy
- 9. Per-Ticker Qualitative Analysis
- 10. What Didn't Work
- 11. Extra Credit
- 12. Limitations & Future Directions
- 13. Reproducibility & Code Structure

1. Methodology & Pipeline Overview

This project builds an end-to-end pipeline that transforms raw earnings-call transcripts into a directional trading signal for 5-day SPY-excess stock returns. The pipeline consists of four stages: (1) transcript parsing and speaker segmentation, (2) dual-LLM structured event extraction, (3) FinBERT speaker-level sentiment scoring, and (4) feature engineering combined with classification modeling. The core thesis is that management tone trajectory and the tension between scripted remarks versus analyst-driven Q&A; carry predictive information about near-term stock direction.

The approach deliberately favors simple, regularized models over complex architectures. With only 131 observations (64 train, 52 test after filtering), overfitting is the dominant risk. Every design decision—from capping tree depth at 2 to applying a dead-zone filter on ambiguous returns—reflects this constraint.

Stage	Input	Method	Output
1. Parse	131 .txt files	Regex + section-header FSM	Structured JSON per call
2. Extract	Prepared + Q&A text	qwen3:14b + gemma3:4b (Anthropic Haiku fallback)	Sentiment, guidance, wins, risks, themes
3. Sentiment	Speaker turns	ProsusAI/finbert	CEO/CFO/analyst scores, prepared vs Q&A slippage
4. Model	Feature matrix (13 cols)	GBM depth=2, 100 estimators	5D up/down prediction

Table 1. Pipeline architecture: four stages from raw transcript to trading signal.

2. Stage 1: Transcript Parsing

Each of the 131 S&P-formatted; transcripts is parsed into a structured representation using a finite-state-machine approach that tracks section headers (Presentation Operator Message, Presenter Speech, Question, Answer). The parser extracts: the company name and call date from the header line; prepared-remarks blocks tagged with speaker role strings; and question-answer pairs matched to analyst and executive speakers.

An upgraded v2 parser was developed to handle edge cases. The original parser collapsed all prepared remarks into a single block, while v2 correctly splits each speaker's prepared segment separately (e.g., AMD Q1-2024: 1 block with v1 vs. 4 blocks with v2). The v2 parser also handles orphan Answer blocks that appear without a preceding Question—a known data quality issue in several transcripts—by creating stub entries rather than dropping content.

Speaker roles are classified into buckets (CEO, CFO, IR, Analyst, Operator, Other) using keyword matching against the role description strings. This classification feeds directly into Stage 3 for role-level sentiment aggregation. The parser processes all 131 transcripts in under 0.2 seconds.

3. Stage 2: Dual-LLM Event Extraction

3.1 Prompt Design

Each transcript is sent to an LLM with a strict-JSON extraction prompt that requests exactly seven fields: overall_sentiment (float in [-1, 1]), sentiment_bucket (5-level ordinal), guidance_direction (raised / reaffirmed / lowered / mixed / none), guidance_confidence (float in [0, 1]), up to 5 wins with severity scores [1–5] and evidence quotes, up to 5 risks with the same structure, and a set of thematic tags. The prompt explicitly requires evidence grounding: each win and risk must cite a passage from the transcript. If uncertain, the model is instructed to lower confidence rather than invent.

Early iterations of the prompt revealed several failure modes. Asking for 'confidence percentage' produced inconsistent formats across outputs—some returned '75%', others '0.75', others 'high'. Risk severity as free text returned entire paragraphs instead of integer scores. Both issues were resolved by switching to strict JSON schemas with explicit enum constraints and numeric ranges. Chain-of-thought reasoning was also tested in prompts but added latency without measurably improving extraction quality, so it was dropped.

3.2 Dual-Model Architecture

The primary extraction model is qwen3:14b via Ollama, with gemma3:4b as a secondary extractor. Both models receive the identical prompt. When Ollama is unavailable (e.g., cloud execution), the pipeline falls back to Anthropic's claude-3-5-haiku-latest, with cost guardrails capping spend at \$2.50 per full run. In the submitted version, the Anthropic backend was used due to the cloud execution environment, with the primary and secondary models set to the same Haiku endpoint.

For each transcript, an **agreement score** is computed between the two extraction outputs. This score combines Jaccard similarity on win labels (30%), risk labels (30%), theme labels (25%), and guidance-direction match (15%), penalized by the absolute gap in overall sentiment. The agreement score serves as a feature in downstream modeling—the hypothesis is that calls where two independent models agree are more clearly bullish or bearish, and thus more predictable.

A forgiving JSON parser handles common LLM output failures: stripping markdown fences, removing <think> blocks (a known issue with Qwen3 and Gemma3), trimming to the outermost braces, fixing trailing commas, and normalizing Python-style booleans. Raw LLM outputs are always saved to disk alongside parsed JSONs to enable debugging. All 131 extractions were successfully cached.

3.3 Extraction Quality: Sample Outputs

To illustrate extraction quality, consider two contrasting cases. **NVDA Q4-2024** (reported Feb 2024) received an overall sentiment of +0.92 (very bullish) with guidance direction 'raised' and confidence 0.85. The top win was 'Exceptional Data Center Growth' (severity 5) citing '\$18.4 billion revenue, up 409% YoY.' The top risk was 'China Market Restrictions' (severity 4). This matches the widely reported blowout quarter.

INTC Q3-2024 (reported Oct 2024) received sentiment of +0.35 (mildly bullish) with guidance 'reaffirmed' and confidence 0.72. Wins included '18A Progress' and 'AWS Foundry Partnership' (severity 3), while risks included 'Gaudi Accelerator Underperformance' (severity 4) with the evidence quote noting the company would miss its \$500M Gaudi revenue target. The nuanced extraction correctly distinguishes between a turnaround-in-progress narrative and a pure earnings beat.

4. Stage 3: FinBERT Speaker-Level Sentiment

In parallel with LLM extraction, ProsusAI/finbert is applied to every speaker turn in every transcript. FinBERT classifies each text chunk as positive, negative, or neutral and returns a confidence score. These are converted to a continuous sentiment score in [-1, 1] by multiplying confidence by sign (positive = +score, negative = -score, neutral = 0).

Scores are aggregated per call along two dimensions. **By role:** sent_ceo (mean over all CEO blocks), sent_cfo (CFO blocks), sent_analyst (analyst question blocks). **By section:** sent_prepared (all prepared remarks) and sent_qa_answer (all executive answers in Q&A;). Each text chunk is truncated to 1,800 characters for FinBERT's 512-token window, and processing is batched in groups of 12 for efficiency.

The key derived feature is **prepared_qa_slippage** = sent_prepared – sent_qa_answer. When this is positive ('reactive'), management tone drops from scripted remarks to unscripted Q&A; answers, suggesting the prepared narrative is more optimistic than executives can defend under analyst questioning. When negative ('proactive'), executives are more positive in Q&A; than in prepared remarks, suggesting genuine confidence. This feature operationalizes the reactive-vs-proactive executive concept from the S&P; research referenced in the assignment.

5. Stage 4: Feature Engineering (13 Features)

The final feature matrix contains 13 features per (ticker, quarter) observation, drawn from four sources: LLM extraction, FinBERT sentiment, dual-LLM agreement, and yfinance price data. All features are constructed with strict no-lookahead discipline: temporal deltas use shift(1) before rolling, and price-context features use only closing prices strictly before the call date. A dedicated audit cell in the notebook programmatically verifies no lookahead violation for mom_21d and dist_52w_high across all 131 rows.

Feature	Source	Description
sentiment	LLM (qwen3)	Overall bullish/bearish score [-1, +1]
sent_ceo	FinBERT	CEO-attributed sentiment (mean over turns)
sent_cfo	FinBERT	CFO-attributed sentiment
sent_analyst	FinBERT	Analyst question sentiment
prepared_qa_slippage	FinBERT	Prepared remarks minus Q&A answer tone
agreement_score	Dual-LLM	Inter-model extraction consistency [0, 1]

Feature	Source	Description
guidance	LLM	raised / reaffirmed / lowered / mixed (categorical)
guidance_confidence	LLM	Extraction certainty [0, 1]
guidance_credibility_gap	LLM	guidance_confidence minus analyst pressure
n_risks / avg_risk_severity	LLM	Risk count and mean severity [1-5]
sentiment_delta	LLM+FinBERT	QoQ change in overall sentiment
guidance_trajectory	LLM	Rolling 3-quarter prior guidance direction sum
mom_21d / dist_52w_high	yfinance	21-day momentum; distance from 52-week high

Table 2. Complete feature set. No look-ahead: all delta/rolling features use shift(1); price features use pre-call window only.

5.1 Feature Selection

A critical lesson from early experiments was the importance of aggressive feature selection. The initial 25-feature set (including ticker one-hot encodings and all flags) achieved strong training IC (~0.40) but collapsed to an OOS IC of just 0.02. Selecting the top 5 features by univariate Spearman IC against the training target improved OOS IC from 0.02 to 0.14—a 7x improvement. The final top-5 feature set used by the best model includes: sentiment, sent_ceo, prepared_qa_slippage, guidance_confidence, and mom_21d. This confirms that with n=64 training samples, parsimony is essential.

6. Target Design & Train/Test Split

6.1 Horizon Selection

Four return horizons were evaluated: +1d, +5d, +21d, and +63d, all computed as SPY-excess returns starting from T+1 close to avoid look-ahead. Initial experiments used regression (ElasticNet) on the 21-day horizon, but this produced R-squared of approximately zero out-of-sample and Spearman IC near zero. The 21-day window is too long—macro drift dominates the earnings-specific signal. Switching to 5-day binary classification (UP/DOWN excess return direction) improved directional accuracy from approximately 50% to 60%, and the 5-day horizon was adopted for all subsequent modeling.

A **dead-zone filter** removes observations where the absolute 5-day excess return is below 0.5%. These near-zero returns are essentially noise and unpredictable; removing them sharpens the signal for both training and evaluation. After filtering, the dataset contains 64 training and 52 test observations.

6.2 Train/Test Split

The split follows the assignment specification: the first 5 calls per ticker form the training set, and all subsequent calls form the test set. This is strictly chronological within each ticker, ensuring no future information leaks into training. The training set spans roughly Q4-2023 through Q1-Q2 2025, and the test set covers Q2-2025 through Q1-2026.

7. Model Comparison (OOS, n=52, 5D Classification)

Eight models were evaluated in a structured progression from simple baselines to ensemble methods. All models use the same chronological train/test split and the same 5-day binary classification target with dead-zone filtering. The table below reports out-of-sample metrics on the 52 test observations.

Model	Hit Rate	Rank IC	ROC-AUC	Sharpe
A: LogReg (25 features)	0.533	0.023	—	—
B: LogReg balanced (top-5)	0.500	0.068	—	—
C: RF depth=2 (top-5)	0.462	0.008	—	—
D: Calibrated LR+RF (top-5)	0.596	0.079	0.554	—
E: GBM depth=2 ★ best	0.596	0.142	0.595	1.08
F: Stacking → soft fallback	0.577	0.112	0.573	—
G: LogReg + guidance_traj	0.577	0.101	0.533	—
H: GBM threshold-tuned (0.50)	0.596	0.142	0.595	—

Table 3. Model comparison. GBM depth=2 (Model E, highlighted) achieves the best Rank IC (0.142) and ROC-AUC (0.595). Sharpe applies to Model E equity simulation only.

7.1 Progression Narrative

Models A–C (baselines): Logistic regression with all 25 features (Model A) overfits badly: train IC of ~0.40 versus OOS IC of 0.023. Balanced class weights (Model B) do not help with such a small sample—hit rate drops to 50% (coin flip). Random Forest with depth=2 (Model C) performs worst at 46.2%, likely because bagging averages away the weak signal.

Model D (Calibrated LR+RF): Combining calibrated logistic regression with random forest via soft voting produces a substantial jump to 59.6% hit rate and 0.079 Rank IC. Calibration helps align probability outputs for effective ensembling.

Model E (GBM, best): Gradient boosting with depth=2, 100 estimators, learning rate 0.05, and subsample=0.8 achieves the best OOS performance. The key advantage of GBM over RF is that it builds trees sequentially, where each tree corrects residuals from the prior ensemble. This allows GBM to capture non-linear interactions (e.g., `n_risks` multiplied by `avg_risk_severity`) without hand-crafting product features. The 0.8 subsample rate adds stochasticity similar to dropout, reducing overfitting.

Model F (Stacking): Stacking GBM and LogReg outputs through a meta-learner LogReg fails. With $n=64$ train and 3-fold CV, each fold has only ~21 training samples. GBM overfits per fold, producing out-of-fold probabilities that are anti-correlated with the target. The meta-learner assigns GBM a negative weight (-0.25)—a clear sign of instability. The pipeline detects this degeneration automatically and falls back to an equal-weight soft vote, which still underperforms standalone GBM.

Model G: Adding the `guidance_trajectory` feature (rolling 3-quarter sum of guidance direction) to LogReg adds +0.025 Rank IC over baseline LogReg, confirming that temporal features carry incremental signal. However, the overall performance remains below GBM.

Model H (Threshold-tuned GBM): Optimizing the classification threshold on the training set yields the same optimal threshold (0.50) as the default, so Models E and H are identical in this case.

8. Backtest: GBM 5D Long/Short Strategy

The backtest simulates a single-name event-driven strategy using Model E predictions. For each of the 52 test-set earnings calls, the strategy takes a long position if the GBM predicts UP and a short position if it predicts DOWN, entering at T+1 close and holding for 5 trading days. Returns are measured as SPY-excess returns to isolate the stock-specific signal from market beta.

Metric	Value
Total trades	52
Hit rate	59.6%
Cumulative excess return	+40.8%
Buy-and-hold benchmark (SPY excess)	+43.0%
Average winning trade	+4.28%
Average losing trade	-4.38%
Profit factor	1.44
Annualized Sharpe ratio	1.08

Table 4. Backtest performance summary for the GBM 5D long/short strategy.

The strategy's cumulative excess return of +40.8% is achieved through consistent directional accuracy rather than outsized wins: the average win (+4.28%) and average loss (-4.38%) are nearly symmetric, so the edge comes from hitting 59.6% of trades correctly. The profit factor of 1.44 means gross winning trades are 1.44 times gross losing trades. The annualized Sharpe of 1.08 is respectable for an event-driven NLP strategy, though the small sample (52 trades) means this estimate carries a wide confidence interval.

The equity curve shows a difficult first half (trades 1–25), where the strategy draws down to approximately -25% cumulative before recovering strongly in late 2025 and early 2026. This pattern suggests the model may benefit from regime-dependent features or a larger training window. The buy-and-hold SPY-excess benchmark, which represents simply holding each stock for 5 days post-earnings without directional prediction, finishes at +43.0%—slightly above the strategy. This illustrates that the strategy's value is in risk management (avoiding large drawdowns from wrong-way bets) rather than raw cumulative return.

9. Per-Ticker Qualitative Analysis

9.1 NVIDIA (NVDA)

NVDA presents the clearest narrative in the dataset: consistently raised guidance and high positive sentiment across every quarter, driven by explosive datacenter revenue growth from AI chip demand. LLM-extracted sentiment ranges from +0.70 to +0.95 throughout the sample, and guidance direction is 'raised' in nearly every call. However, the pipeline reveals an important subtlety: while the NLP signal remains strongly bullish, 5-day excess returns diverged after Q4-2024 as the stock's valuation stretched. High sentiment did not reliably predict positive excess returns when expectations were already fully priced in. This illustrates a fundamental limitation of sentiment-based signals: they capture the narrative but not the market's prior expectations.

9.2 Intel (INTC)

INTC provides the inverse case: persistent guidance cuts and declining sentiment over most of the sample. The LLM correctly flags recurring risks like 'Gaudi underperformance' and 'gross margin pressure' across multiple consecutive quarters, while wins are qualified and uncertain (e.g., 18A process progress, AWS partnership). The pipeline consistently predicted DOWN for INTC, and the hit rate for this ticker was above the portfolio average. However, magnitude varied significantly—some negative earnings calls were followed by small excess losses while others triggered large drawdowns, highlighting the difference between direction and magnitude prediction.

9.3 FedEx (FDX)

FDX demonstrates the value of temporal features. The company went through a recovery arc: guidance was consistently cut throughout 2024, then gradually rebuilt during 2025 as the DRIVE transformation program gained traction. The sentiment-delta feature captured this inflection—the shift from negative to positive quarter-over-quarter changes—and the model's prediction accuracy for FDX improved notably from Q2-2025 onward as the trajectory feature began reflecting the turn. The LLM also tracked the emergence of the FedEx-spinoff theme (network optimization) as a novel positive, which was absent from earlier calls.

10. What Didn't Work

10.1 Stacking Ensemble (Model F)

Stacking GBM and LogReg through a meta-learner failed because the out-of-fold predictions from 3-fold CV on $n=64$ were too noisy. Each fold had only ~21 training samples, causing GBM to overfit within each fold. The resulting OOF probabilities were anti-correlated with the true target, leading the meta-learner to assign GBM a negative weight (-0.25). This is a well-known pathology: stacking generally requires 300+ samples to produce stable OOF estimates. The pipeline detected this degeneration and fell back to soft voting automatically.

10.2 Full Feature Set (25 Features)

The initial model with all 25 features (including ticker one-hot encodings, all extraction flags, and interaction terms) appeared strong in training (IC ~0.40) but generalized poorly (OOS IC 0.02). This is a textbook demonstration of the bias-variance tradeoff on small samples. Feature selection to the top 5 by univariate Spearman IC improved OOS IC by 7x (0.02 to 0.14). The lesson is clear: with small datasets, fewer, higher-quality features dominate.

10.3 Regression on 21-Day Returns

ElasticNet regression on 21-day excess returns produced R-squared of approximately zero and Spearman IC of approximately zero out-of-sample. The 21-day window absorbs too much macro drift (interest rate moves, sector rotation, geopolitical events) that is unrelated to the earnings call content. The 5-day horizon retains enough of the post-earnings drift while limiting macro contamination. Switching from regression to binary classification also helped by eliminating the need to predict magnitude, which is extremely noisy at the single-stock level.

10.4 Cross-Sectional Long/Short

Only 4 test quarters had 6 or more concurrent reporters (the minimum needed for a meaningful top-3/bottom-3 long/short portfolio). The hit rate was just 25% (1 of 4 quarters positive) with cumulative return of -2.4%. The sample is far too small for this approach to be evaluated fairly. Single-name long/short

(52 trades, Sharpe 1.08) proved much more reliable at this data volume.

10.5 Prompt Engineering Failures

Several prompt design decisions were iterated on before finding a stable configuration. Asking for 'confidence percentage' produced inconsistent output formats across calls (75%, 0.75, 'high'). Risk severity requested as free text returned entire paragraphs instead of scores. Both were fixed by specifying strict JSON with explicit enum and numeric range constraints. Additionally, chain-of-thought prompting was tested but added ~40% latency per call without improving extraction F1, so it was dropped in favor of direct extraction.

11. Extra Credit

11.1 Speaker-Level Sentiment Analysis

FinBERT was applied per speaker turn to produce separate sentiment features for CEO, CFO, and analyst roles. The speaker-sentiment-vs-return analysis reveals a notable pattern: CEO sentiment shows a slight inverse relationship with 5D returns at the highest sentiment bins (very bullish CEOs are associated with average returns near zero), while CFO sentiment shows a positive monotonic relationship. Analyst sentiment, measured from question tone, shows a U-shape: very negative analyst questions are associated with positive subsequent returns (contrarian signal), and very positive analyst tone also correlates with positive returns. The CEO bullish-sentiment finding is consistent with the hypothesis that overly promotional CEO language may be a contrary indicator.

11.2 Two-LLM Comparison

The dual-model architecture (qwen3:14b primary + gemma3:4b secondary) enables computing an agreement score per extraction. The agreement score is used as a feature in the model and also evaluated directly as a signal filter. The analysis shows that high-agreement calls (where both models converge on similar extractions) achieve 67% directional accuracy, while low-agreement calls (n=9) drop to 44% accuracy. Medium-agreement calls achieve 60%. This confirms the intuition that model consensus is a proxy for call clarity—ambiguous calls are harder to predict because the fundamental signal is weaker.

11.3 External Signal Integration

Two price-based features were integrated from yfinance: 21-day trailing momentum (mom_21d) and distance from 52-week high (dist_52w_high). Both use prices strictly before the call date to prevent lookahead. Momentum combined with the NLP features in the top-5 feature set of the GBM model, confirming that price context adds incremental information beyond text analysis. The anti-lookahead property was programmatically verified across all 131 rows.

11.4 Reactive-vs-Proactive Executive Analysis

The prepared_qa_slippage feature operationalizes the S&P; research concept of reactive versus proactive executives. When Q&A-answer; sentiment exceeds prepared-remarks sentiment (proactive), the average 5D excess return is +1.4% (n=37). When prepared-remarks sentiment exceeds Q&A-answer; sentiment (reactive), the average return is +0.7% (n=78). The Spearman correlation between slippage and return is -0.117, indicating that proactive executives (more positive in Q&A; than in script) tend to outperform. While the effect is modest and the sample is small, the direction is consistent with the theory that executives who can be more enthusiastic spontaneously under questioning are signaling genuine confidence.

11.5 Cross-Sectional Long/Short

A cross-sectional strategy was tested: each quarter, rank all reporting tickers by model signal, go long the top 3, and short the bottom 3. Only 4 test quarters had sufficient concurrent reporters. The results were poor: 1 of 4 quarters positive, cumulative return of -2.4%. This is not statistically meaningful with only 4 data points, and confirms the assignment's observation that cross-sectional strategies require far more breadth (e.g., S&P; 500 with 4,000+ events/year) to be reliable.

Extra Credit Item	Status	Key Finding
Speaker-level sentiment	Done	CEO bullish sentiment has contrarian signal; CFO sentiment is monotonically predictive
Two-LLM comparison	Done	High-agreement calls: 67% accuracy vs. low-agreement: 44%
External signal integration	Done	mom_21d in top-5 feature set; anti-lookahead verified
Reactive-vs-proactive	Done	Proactive execs: +1.4% avg return vs. reactive: +0.7% ($\rho = -0.117$)
Cross-sectional L/S	Limited	4 quarters only, cum -2.4%; not statistically reliable
Prompt ablation	Planned	Would require 50 hand-labeled calls; deferred due to annotation cost

Table 5. Extra credit summary.

12. Limitations & Future Directions

12.1 Limitations

Small sample size. With $n=130$ total observations (52 OOS), most statistical conclusions carry wide confidence intervals. The 59.6% hit rate has a standard error of approximately 7%, meaning the true hit rate could plausibly be anywhere from 52% to 67%. The Sharpe ratio of 1.08 is estimated from only 52 non-overlapping trades, which is insufficient for reliable Sharpe estimation.

Narrow universe. The 14-ticker universe is concentrated in technology and financials. Results may not generalize to consumer staples, utilities, energy, or other sectors where earnings-call dynamics differ materially. Sector effects could also be confounded with the NLP signal.

LLM extraction quality. Without manually annotated ground-truth labels for sentiment, guidance direction, and risk severity, extraction quality cannot be rigorously evaluated. The dual-LLM agreement score provides a proxy, but it measures model consistency rather than accuracy.

Macro noise in returns. Even the 5-day return window includes macro shocks unrelated to the earnings call (e.g., Fed announcements, geopolitical events). The dead-zone filter helps but does not eliminate this issue.

12.2 With More Data

Expanding to S&P; 500 coverage (4,000+ earnings events per year) would enable reliable cross-sectional long/short strategies, proper sector-neutral evaluation, and walk-forward optimization with enough statistical power. Using 10-Q filings to cross-validate extracted risks against regulatory disclosures would provide ground-truth for extraction quality. Incorporating analyst estimate revisions post-call would add a stronger fundamental signal.

12.3 With More Time

Prompt-engineering ablation: Hand-label 50 calls and compare zero-shot vs. 3-shot vs. chain-of-thought extraction on field-level F1 for sentiment, guidance, and severity separately. **Longer-horizon combination:**

Use the 5-day signal for entry timing combined with a 63-day signal for position sizing, as 63-day excess returns show a cleaner relationship with guidance trajectory. **Continuous signal tracking:** Track sell-side note sentiment between earnings calls and apply as a decay factor: a bullish call loses signal if subsequent coverage turns bearish. **Better cross-sectional strategy:** With 5+ years of data, rank within sector and combine with macro factor hedges.

13. Reproducibility & Code Structure

The submission is fully reproducible without API calls. All 131 LLM extraction JSONs and FinBERT sentiment outputs are cached in SUBMISSION_CACHE/. The pipeline can be re-run from cached artifacts in approximately 2 minutes (parsing + feature construction + model training + evaluation).

File / Directory	Purpose
Pakala_Chaithanya_MTH9796_HW1.ipynb	Main notebook: full pipeline from parse to backtest
src/earnings_pipeline/parser.py	Transcript parser (v1 + v2 with speaker turns)
src/earnings_pipeline/extraction.py	Dual-LLM extraction engine, agreement scoring, JSON repair
src/earnings_pipeline/sentiment.py	FinBERT speaker-level sentiment (role bucketing, batching)
src/earnings_pipeline/features.py	Feature construction with anti-lookahead price features
src/earnings_pipeline/config.py	Model names, API endpoints, context window settings
SUBMISSION_CACHE/extractions/	131 cached extraction JSONs (no API cost to re-run)
SUBMISSION_CACHE/features_*.parquet	Pre-built feature tables for all horizons
cache/model_results_*.csv	Saved model evaluation metrics
ECT/	131 raw earnings-call transcripts (.txt)

Table 6. Repository structure.

Dependencies: pandas, numpy, yfinance, scikit-learn, transformers (for FinBERT), requests, tqdm, matplotlib. Optional: ollama (for local LLM), anthropic SDK (for API extraction). All cached artifacts are included so the modeling and evaluation stages can be reproduced without any LLM or API access.

End of writeup. Total: 131 transcripts processed, 13 features engineered, 8 models evaluated, 5 extra credit items completed. Best model: GBM depth=2, Accuracy 59.6%, Rank IC 0.142, Sharpe 1.08.